Designing a 4-bit Adder/Subtractor

with Overflow Detection

Working, Design, and Applications

DDCA PROJECT

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INTRODUCTION

One compact circuit performs two operations—and knows when a signed result can no longer be trusted.

01

WHY

Computers, calculators, and embedded systems depend on fast binary arithmetic.

02

WHAT

A 4-bit datapath shares four full adders for addition and subtraction.

03

CHECK

Overflow logic protects signed results from silent wrap-around.

04

SCOPE

Architecture, working modes, overflow detection, and applications.

Basics of Binary Addition & Subtraction

- **Signed numbers:** MSB represents the sign (0 = positive, 1 = negative).
- **Overflow:** Occurs when the result of an operation exceeds the representable range.
  - For 4-bit signed numbers: range is **-8 to +7**

ADD

```
0+0=0    0+1=1
1+0=1    1+1=10
```

SUBTRACT

$$A - B = A + \bar{B} + 1$$

SIGNED RANGE

```
1000   …   0111
 −8         +7
```

Design Overview – Block Diagram



-
  -
  -
  -
    -
    -

Core components:

4 Full Adders (FA) connected in ripple-carry configuration.

4 XOR gates (one per B input) to conditionally invert B.

A control signal **SUB**:

SUB = 0 → Addition (A + B)

SUB = 1 → Subtraction (A - B)

- **Overflow detection logic:** XOR of carry-in and carry-out of the MSB full adder.
- **Diagram suggestion:** Draw a 4-bit ripple carry adder, with each B input passing through an XOR gate whose other input is SUB. The SUB signal also feeds the carry-in of the LSB.

Full adder: Building Block

- Inputs: A, B, Cin
- Outputs: Sum, Cout

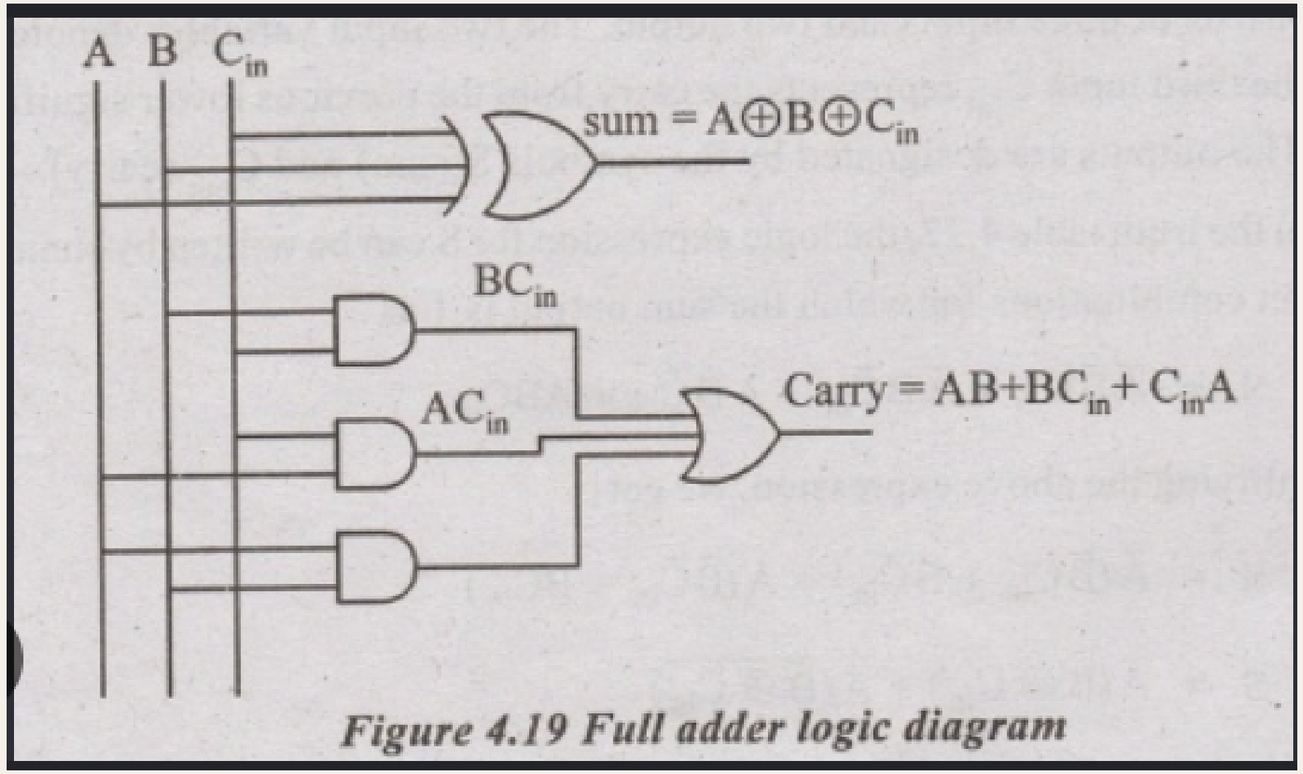


Figure 4.19 Full adder logic diagram

TRUTH TABLE-

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

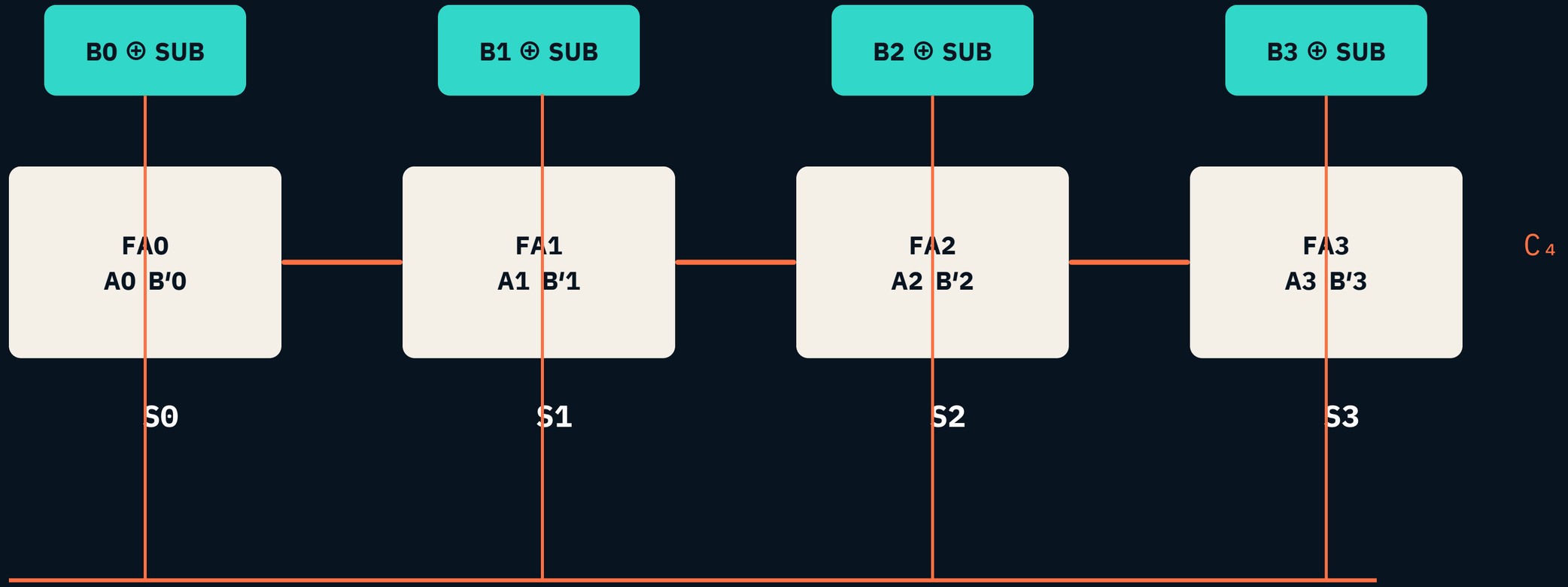
$$Sum = A \oplus B \oplus Cin$$

$$Cout = (A \cdot B) + Cin \cdot (A \oplus B)$$

The 4-bit ripple-carry circuit

SUB fans out to every B XOR and seeds C_0.

4 Full Adders: FA0 (LSB) to FA3 (MSB).
XOR gates: Each B input Bi*Bi* is XORed with SUB.
If SUB = 0 → Bi⊕0=Bi*Bi*⊕0=*Bi* (addition)
If SUB = 1 → Bi⊕1=Bi¯*Bi*⊕1=*Bi* (subtraction)
Carry-in of FA0: Connected to SUB.
For addition: Cin = 0
For subtraction: Cin = 1 (completes two's complement)
Outputs: S0–S3 (Sum/Difference), C4 (final carry out).



Working – Addition Mode (SUB = 0)

SUB = 0 • C_0 = 0

```
  0010    (+2)
+ 0011    (+3)
------------------
  0101    (+5)
```

- SUB = 0
- B inputs pass unchanged: Bi⊕0=Bi
- LSB carry-in = 0
- Circuit performs: A + B + 0
- Example: A = 0010 (+2), B = 0011 (+3)
  - Sum = 0101 (+5) → No overflow (C3 = 0, C4 = 0)
- Result: Correct sum with carry out C4.

MATHEMATICAL RESULT

+5

inside 4-bit signed range so v = 0

$C_3 = C_4$

V = 0

+5 lies inside −8...+7. Equal carries around the sign bit mean no overflow.

Subtraction is addition in disguise

SUB = 1 → invert B → C0 = 1

$$\begin{aligned} B &= 0011 \\ \bar{B} &= 1100 \\ \bar{B} + 1 &= 1101 \end{aligned}$$

$$A + \bar{B} + 1 = A - B$$

Carry into and out of the MSB are equal, so V = 0. The result +2 is correct.

```
0101 + 1101 = 1 0010
discard C4  →   0010 (+2)
```

- **SUB = 1**
- **B inputs inverted: Bi⊕1=Bi’**
- **LSB carry-in = 1**
- **Circuit performs: A + B' + 1 = A - B**
- **Example: A = 0101 (+5), B = 0011 (+3)**
- **B' = 1100, +1 -> 1101 (-3 in two's complement)**
- **0101 + 1101 = 10010 -> discard carry, result = 0010 (+2) -> Correct**
- **Overflow check: C3 = 1, C4 = 1 → No overflow.**

Overflow Detection – Concept

positive + positive

negative result

negative + negative

positive result

$$V = C_3 \oplus C_4$$

Overflow occurs when:

Adding two positive numbers gives a negative result.

Adding two negative numbers gives a positive result.

For addition/subtraction circuit:

Let C3 = carry into MSB (from FA2 to FA3)

Let C4 = carry out of MSB (from FA3)

Overflow $V = C3 \oplus C4$

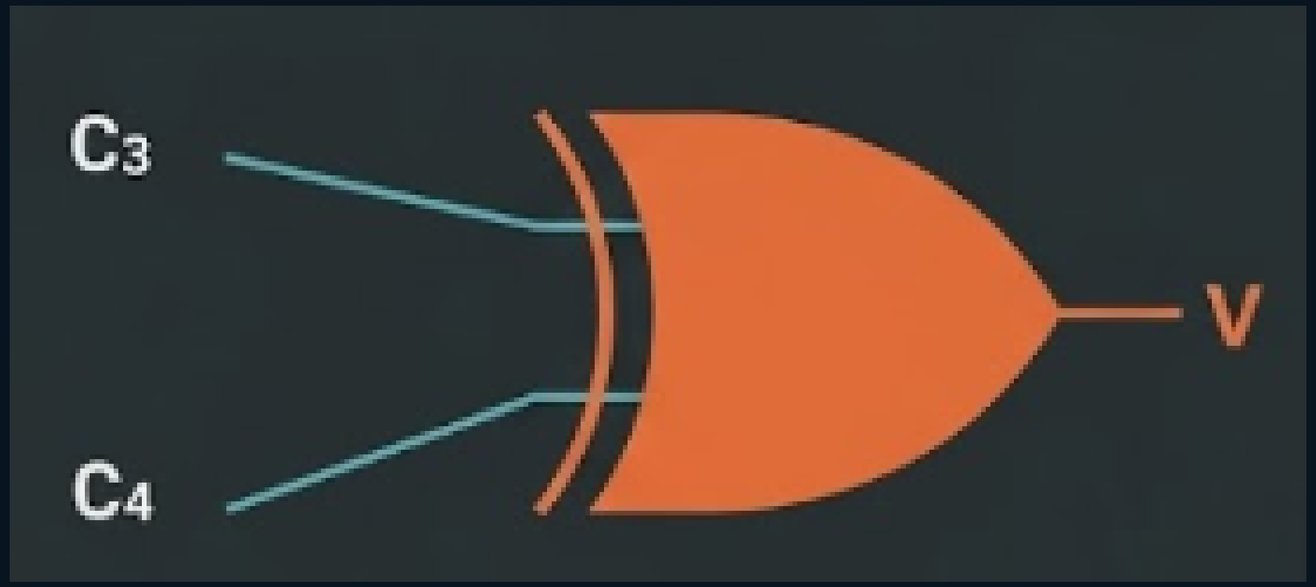
V = 1 → Overflow

V = 0 → No overflow

This works for both addition and subtraction because subtraction is treated as addition of two's complement.

Two carries decide the flag

- **Circuit:** XOR gate with inputs C3 and C4. Output V.



C_3	C_4	V
0	0	0
0	1	1
1	0	1
1	1	0

0111 (+7) + 0001 (+1) = 1000 (−8)
C_3=1, C_4=0 → V=1

- **Example (Overflow):** A = 0111 (+7), B = 0001 (+1) → Sum = 1000 (-8) → C3=1, C4=0 → V=1.

Example (No Overflow): A = 0010 (+2), B = 0011 (+3) → Sum = 0101 (+5) → C3=0 → C4=0 →V=0

Applications

ALU

The arithmetic core of processors and microcontrollers.

DSP
filtering & FFT

CALCULATORS
basic computing

EMBEDDED
robotics & automotive

FPGA / ASIC
custom arithmetic

SAFETY
avionics & medical

CONTROL
real-time systems

Conclusion

01 **SHARE**

XOR gates and SUB let one datapath add or subtract.

02 **DETECT**

$C_3 \oplus C_4$ exposes signed overflow immediately.

03 **SCALE**

The ripple structure extends naturally to wider words.

- A 4-bit adder/subtractor can be built using 4 full adders and 4 XOR gates.
- The **SUB** control signal selects addition or subtraction.
- **Overflow detection** uses $V=C3 \oplus C4$.
- This design is modular, scalable (to 8-bit, 16-bit, etc.), and widely used in digital systems.
- Understanding this circuit is essential for anyone studying computer architecture or digital logic design.

A foundational circuit for digital logic and computer architecture.